



Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no.....

Name of the Course: B.Tech CSE
Semester: IV
Name of the Paper: Microprocessors
Paper Code: TCS-403
Time: 1.5 hour

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub-questions
- (ii) Each question carries 10 marks.

Q1. (10 Marks)

a. Explain the **evolution of microprocessors** and describe the basic structure of a **microcomputer system** with a neat diagram. CO 1

OR

b. Describe the **internal architecture of the 8085 microprocessor** and explain the function of each block. CO 1

Q2. (10 Marks)

a. Draw and explain the **pin diagram of 8085**. Discuss the function of all pins in detail. CO1

OR

b. What are address bus, data bus, and control bus? Explain their role in a microprocessor-based system. CO1

Q3. (10 Marks)

a. Explain the memory interfacing technique of 8085. Design a memory interface for 4 KB RAM with necessary logic. CO2

OR

b. Describe the instruction set of 8085 and classify instructions with suitable examples. CO2

Q4. (10 Marks)

a. Explain different addressing modes of 8085 with examples. CO2

OR

b. Write an assembly language program in 8085 to find the largest number in a given array. CO3

Q5. (10 Marks)

a. Explain the **interrupt structure of 8085**. Discuss maskable and non-maskable interrupts. CO2

OR

b. Explain the **machine cycle and instruction cycle of 8085** with timing diagrams. CO3